

E-Banking

Lesson 39 — PINs, OTPs and Safe Payments

JKSSB Computer Awareness

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What E-Banking Means

- ▶ **E-banking** means using banking services through the internet instead of visiting the branch.
- ▶ **Net banking** is banking through the bank's website on a browser; **mobile banking** is banking through the bank's app.
- ▶ **UPI** — Unified Payments Interface — apps move money instantly between bank accounts using a phone.
- ▶ ATM, net banking, mobile banking, and UPI are all channels; the **account stays in the bank**, only the access channel changes.

The one idea

E-banking is remote access to the same bank account through a website, an app, or UPI.

UPI Basics

- ▶ **UPI** stands for Unified Payments Interface.
- ▶ A **VPA** (Virtual Payment Address, e.g. name@bank) identifies the account without revealing the account number.
- ▶ UPI transfers are **instant** and work **24 × 7**, including Sundays and holidays.
- ▶ Every UPI payment is confirmed with a **UPI PIN**; without it the money does not move.

Trap alert

UPI needs the UPI PIN to **send** money — receiving money needs no PIN at all.

ATM PIN vs UPI PIN

- ▶ **ATM PIN** (Personal Identification Number) is entered at an ATM or shop machine to use the **physical card**.
- ▶ **UPI PIN** is entered in the phone to authorise a **UPI payment**; it is set separately for each bank account in the UPI app.
- ▶ The two PINs **can be different** and are used on different channels — ATM PIN never authorises a UPI payment.
- ▶ Both are **secret** and are never asked for by the bank.

Keep the pair straight

ATM PIN = card channel; UPI PIN = UPI channel. Do not swap them.

CVV vs OTP

- ▶ **CVV** — Card Verification Value — is the 3-digit number on the **back of the card** (4 digits on the front for some cards).
- ▶ CVV is used with the card number and expiry date for **online card payments**; it proves the card is in hand.
- ▶ **OTP** — One-Time Password — is a short code sent by **SMS** that is valid for **one transaction only** and a few minutes.
- ▶ CVV is **printed and reusable**; OTP is **generated fresh** each time and then expires.

Trap alert

CVV is fixed on the card; OTP changes every time. Never share either.

The Four Codes Side by Side

Code	Where it lives	What it does
ATM PIN	In your memory	Authorises card use at ATM/shop
UPI PIN	In your memory	Authorises a UPI payment
CVV	Printed on the card	Proves card possession online
OTP	Sent by SMS	Approves one transaction, then expires

Exam depth: entering any PIN means you are paying; OTP is single-use.

Authentication and MFA

- ▶ **Authentication** means proving “I am the account holder” before the bank acts.
- ▶ The three factors are **something you know** (password, PIN), **something you have** (phone, card, OTP on SMS), and **something you are** (fingerprint, face).
- ▶ **MFA** — Multi-Factor Authentication — checks **two or more** different factors together.
- ▶ A login password **plus** an SMS OTP is the everyday MFA example: password (know) + phone (have).

The one idea

One factor can be stolen; two different factors together are much harder to steal.

QR Code Payments

- ▶ A **QR** (Quick Response) code is the square barcode the shop displays; scanning it fills in the merchant's UPI address.
- ▶ Always **verify the merchant name and amount** on your own screen before entering the UPI PIN.
- ▶ A QR never asks for your PIN or OTP by itself — the **phone screen** asks, after you verify the payee.
- ▶ A QR sticker **pasted over** the original code is a classic fraud; confirm the displayed name matches the shop.

Office canteen example

The canteen shows one QR; the phone displays “Canteen Stores Rs. 50”. Only that verified screen is approved with the UPI PIN.

Collect Requests: The Key Safety Rule

- ▶ A **collect request** is a message asking **you** to pay someone — it appears in the UPI app as “pay Rs. X to Y”.
- ▶ Approving it and entering your UPI PIN **sends your money**; there is no “receive” step that needs a PIN.
- ▶ **Decline or ignore** every collect request you did not expect, even if the remark says “refund” or “cashback”.
- ▶ To **receive** money, share your VPA or QR; never enter a PIN.

Trap alert

Entering the UPI PIN always means paying. No refund or prize ever needs your UPI PIN.

Safe-Banking Rules I

- ▶ The bank **never** asks for OTP, CVV, ATM PIN, or UPI PIN by call, SMS, or link — any such ask is fraud.
- ▶ **Never share** an OTP to “cancel”, “verify”, or “receive” a payment; sharing it completes the fraud.
- ▶ Do not click **KYC-update or account-blocked** links; open the bank’s site or app yourself by typing its address.
- ▶ **Never install** a screen-sharing app at a caller’s request; it shows the fraudster your OTPs and PINs.

Safe-Banking Rules II

- ▶ Use **typed addresses and official apps** from the app store; check for the **padlock (https)** on the banking site.
- ▶ Avoid **public Wi-Fi** for banking; prefer mobile data or a trusted network.
- ▶ **Log out** after net banking and clear a shared browser; lock the UPI app with the device lock.
- ▶ Keep **SMS and email alerts on** and check the statements, so an unknown debit is noticed at once.

Habit to keep

Pause, verify on your own screen, then approve — never approve in a hurry on a caller's instructions.

Online Transfer Modes in One Line Each

Mode	One-line use
UPI	Instant small payments by phone, 24×7 , authorised by UPI PIN.
IMPS	Immediate Payment Service — instant bank-to-bank transfer, 24×7 .
NEFT	National Electronic Funds Transfer — batched settlement, no minimum amount.
RTGS	Real Time Gross Settlement — large-value, instant, with a minimum amount.

UPI and IMPS are instant; NEFT settles in batches; RTGS is for large values.

Full-Form Ladder

- ▶ UPI — **Unified Payments Interface**
- ▶ OTP — **One-Time Password**
- ▶ CVV — **Card Verification Value**
- ▶ PIN — **Personal Identification Number**
- ▶ QR — **Quick Response** code
- ▶ MFA — **Multi-Factor Authentication**

Recall drill

Cover the right column; UPI, OTP, CVV, and PIN expansions are direct one-mark items.

Fraud Patterns to Recognise

- ▶ **Fake refund/cashback** collect request asking for UPI PIN approval.
- ▶ **KYC/account-blocked** SMS with a link that harvests card details and OTP.
- ▶ **Screen-share** request (“install this support app”) during a bank-sounding call.
- ▶ The response to all three is the same: **stop, do not share any code, and verify through the official app or branch.**

Trap Alert: PIN, CVV and OTP Facts to Keep Straight

- ▶ ATM PIN is for the **card**; UPI PIN is for **UPI** — they are not interchangeable.
- ▶ CVV is **printed on the card** and reusable; OTP is **sent fresh** for one transaction and expires.
- ▶ Entering the UPI PIN **always pays**; receiving money needs **no PIN**.
- ▶ Password + OTP is **two-factor** authentication, not one.
- ▶ The bank **never asks** for OTP, CVV, or any PIN.

Lesson 39: Recall

- ▶ E-banking = net banking, mobile banking, and UPI on the same bank account.
- ▶ UPI (Unified Payments Interface) pays instantly, 24×7 , by VPA; each payment needs the UPI PIN.
- ▶ ATM PIN is the card secret; UPI PIN is the UPI secret; CVV is printed on the card; OTP is single-use over SMS.
- ▶ MFA combines two or more factors: know, have, and are.
- ▶ Scan QR only after verifying name and amount; decline unknown collect requests.
- ▶ Never share OTP/CVV/PIN; the bank never asks; avoid links, screen-share apps, and public Wi-Fi.